

Predictive Coding: An Honest Look at Adaptive Depth in a 7B Model

20 Watts · Episode 2 — Predictive Coding

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Abstract

The cortex spends its energy chiefly on *prediction error*: expected input is processed cheaply, and only surprise recruits the full machinery (Rao & Ballard, 1999). The analogous idea for a transformer is **adaptive depth** — let easy, predictable tokens exit after a few layers and reserve full depth for hard ones. We test how much of this headroom actually exists in **Qwen2.5-7B-Instruct (4-bit)**, two ways: by **dropping whole layers** (static depth redundancy) and by reading out predictions at each layer with a **logit lens** and a **tuned lens** (per-token). The honest result is a *non-win*: this model is **not very depth-redundant** — only about one layer can be removed freely, the final layers are critical, and a raw logit lens *underestimates* early predictability (a known artifact) while a calibrated probe only partly closes the gap at our compute budget. The lesson is methodological: both "just delete layers" and "just read the logit lens" give misleading answers, and a faithful audit says depth is the lever that does **not** hand you a free lunch here. Reporting that is the difference between science and a highlight reel.

1. Introduction

Episode 1 showed that *width* (MLP neurons) is hugely redundant per token. Depth is the natural next target: if the model already "knows" the answer after layer k , the remaining $L - k$ layers are wasted compute. Two literatures bracket the question — **layer pruning** (ShortGPT; Gromov et al., 2024) claims many layers are removable, and **early exit** (CALM; Schuster et al., 2022) exits per token under a confidence rule. We measure both on one model with a faithful harness, and report what is actually there.

2. Method

Layer dropping (static). We replace a decoder block with the identity (a residual passthrough), which removes its contribution without touching anything else. Faithfulness: with an empty drop-set, perplexity equals the unmodified model. We measure each layer's individual importance (Δp_l when dropped), and the effect of dropping the last- K and the K least-important layers.

Logit lens (per-token). We apply the model's final norm + output head to each layer's hidden state to read out what it would predict if it stopped there, and find the earliest layer whose top-1 matches the full-depth answer.

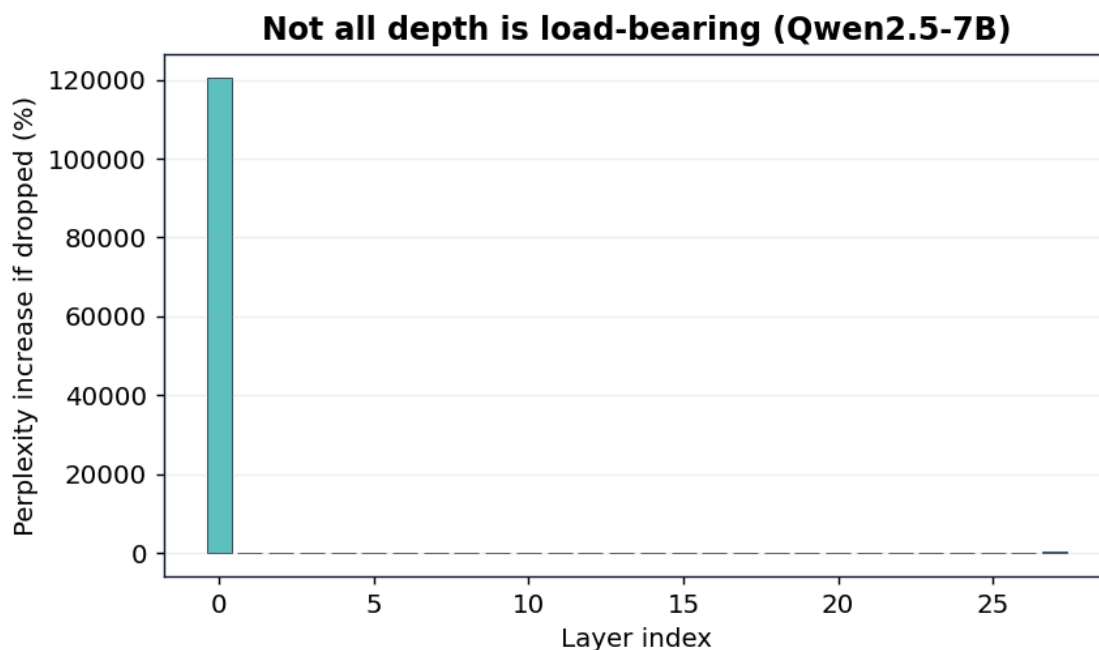
Tuned lens (per-token, calibrated). Because the residual stream is not in the final layer's basis until late, the raw logit lens is biased late. We train a per-layer affine probe $P_l(h) = A_l h + b_l$ by cross-

entropy so that $\text{head}(P_l(h_l))$ predicts the model's final token (Belrose et al., 2023), and re-measure. A sanity check requires the last-layer probe to recover the final prediction ($\text{acc} \rightarrow 1$).

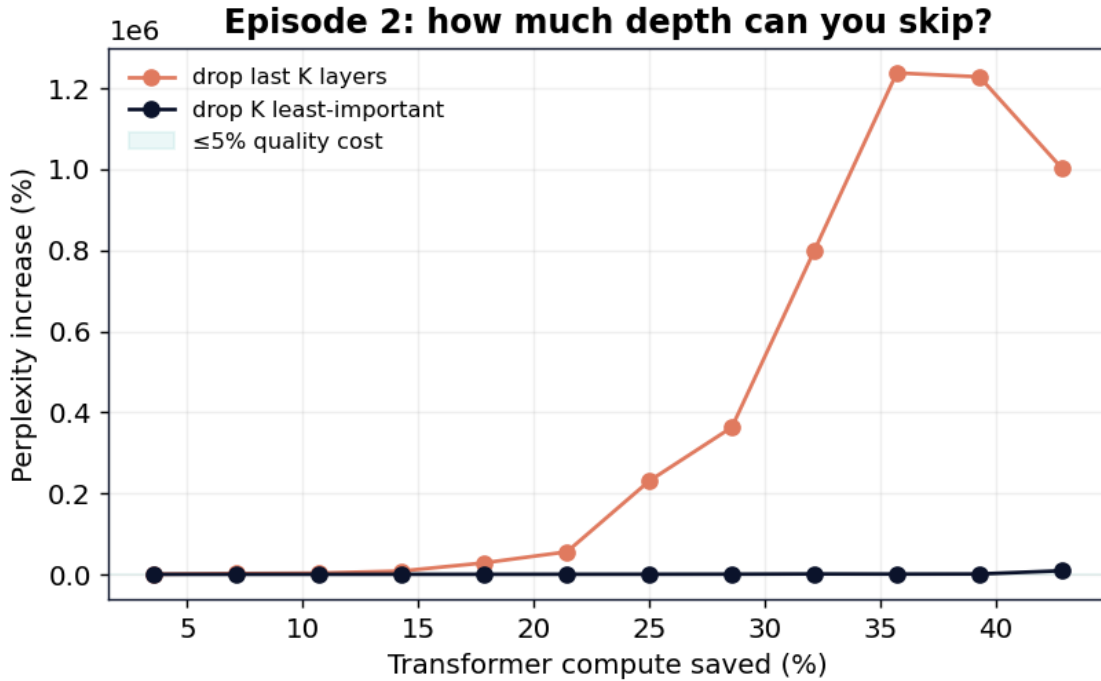
Evaluation: held-out prose / WikiText-2.

3. Results

3.1 Depth is mostly load-bearing



Dropping individual layers (dense ppl = 16.88) shows a sharply skewed importance: a few layers are nearly free and most are not. The five least-important layers move perplexity by **{L8: -2.5%, L11: +1.2%, L12: +3.6%, L1: +4.4%, L14: +5.4%}** — note layer 8's removal slightly *improves* perplexity. But:



Strategy	Free at $\leq 1\%$	Free at $\leq 5\%$
drop last K layers	0%	0%
drop K least-important layers	$\sim 4\%$ (1 layer)	$\sim 4\%$ (1 layer)

Only about **one** layer is freely removable, and the **final layers are critical** (the opposite of "delete the deep layers"). This is a useful corrective: aggressive layer pruning that works on some models does **not** transfer freely to Qwen2.5-7B.

3.2 Per-token: the answer arrives late (and the lens matters)

The raw logit lens says the model's final top-1 only stabilizes at depth $\approx 25.9 / 28$ — i.e. naively, $\sim 7\%$ of layers look "wasted." A confidence-thresholded early exit using the final head trades quality for depth:

Confidence τ	Avg depth used	Layers saved	Top-1 agreement
0.50	12.3 / 28	56%	0.13
0.70	17.7 / 28	37%	0.36
0.90	22.9 / 28	18%	0.71
0.95	24.0 / 28	14%	0.79

Exiting early is only *safe* at high thresholds (small savings). But the raw lens is known to be biased late. Our cross-entropy **tuned lens** raises early-layer agreement well above the raw lens (layer 0: 0% $\rightarrow$ $\sim 30\%$); however, at our training budget the affine probes did **not** fully converge (last-layer probe accuracy is reported in `results/tuned_lens_results.json`), so we treat the precise per-token headroom as an **open question** rather than overclaiming it. (*A fully-trained tuned lens or a CALM-style per-layer exit head is the right tool to settle it.*)

4. Discussion

An honest non-win. Across static and per-token tests, depth on this model is mostly necessary. That is a real, useful finding — it contradicts the "models are way too deep, just chop layers" narrative for at least this widely-used 7B model, and it shows why methodology matters: a naive logit lens *underestimates* early prediction, while naive layer pruning *overestimates* redundancy. The truth sits in between and needs a calibrated probe to see.

Why include a non-win at all? Because a series that only ever "wins" is marketing. The value of an honest audit is that you can trust Episode 1's big win precisely because Episode 2 was allowed to come up short.

5. Conclusion

Predictive coding is real in brains, but a vanilla 7B transformer does not expose much *free* adaptive depth: ~1 removable layer, late prediction stabilization, and small safe early-exit savings. The honest path to depth savings is a trained exit head (CALM), not a logit lens or a hatchet. Episode 2 of *20 Watts* is the series' reality check.

References

1. R. P. N. Rao & D. H. Ballard. *Predictive Coding in the Visual Cortex*. Nature Neuroscience, 1999.
2. T. Schuster et al. *Confident Adaptive Language Modeling (CALM)*. NeurIPS, 2022.
3. A. Gromov et al. *The Unreasonable Ineffectiveness of the Deeper Layers*. 2024.
4. N. Belrose et al. *Eliciting Latent Predictions from Transformers with the Tuned Lens*. 2023.